

Bhavdeep Singh

bhavdeepmakkar@gmail.com · +91-9463448633 · LinkedIn · GitHub

EDUCATION

Indian Institute of Information Technology, Una

2024 – 2028

B.Tech in Information Technology

TECHNICAL SKILLS

- **Languages:** C++, Python, JavaScript
- **Frontend:** React.js, HTML, CSS, Tailwind CSS
- **Backend:** Node.js, Express.js, Flask, REST APIs, JWT Authentication
- **Databases:** MongoDB, SQL, Schema Design, Indexing
- **GenAI / ML:** LLM APIs, RAG, Prompt Engineering, AI Integration
- **Deployment:** Vercel, Render, Railway, AWS(EC2, S3)
- **Tools:** Git, GitHub, Postman, Gmail API
- **Concepts:** DSA, OOP, DBMS, OS, Computer Networks

PROJECTS

Opportuneo – AI-Powered Job Outreach SaaS *FastAPI, React, Gemini, Groq, Gmail API, OAuth 2.0, RabbitMQ, Supabase, Railway, Razorpay* www.opportuneo.in

- Built and deployed a full-stack SaaS platform integrating 8+ external services/APIs including Gmail API, Google OAuth, Gemini, Groq, RabbitMQ, Supabase, Razorpay, and Railway.
- Implemented a credit-based outreach system capable of generating and sending 1,000+ personalized emails per subscription plan directly through users' connected Gmail accounts.
- Designed an asynchronous processing architecture using RabbitMQ with retry handling and dead-letter queues, reducing user-facing wait times by offloading long-running email campaigns to background workers.
- Developed 15+ production REST APIs covering authentication, payments, subscriptions, email generation, campaign execution, resume management, and Gmail integration. ““

Finance Tracker — MERN Stack

MongoDB, Express.js, React.js, Node.js, JWT, Tailwind CSS

GitHub

- Delivered a full-stack budgeting platform tracking 5+ financial categories with real-time trend charts and monthly summaries.
- Locked down all 12 API routes using JWT auth and role-based access, eliminating cross-user data exposure.
- Introduced compound indexes on MongoDB collections, slashing average query time from ~120ms to ~90ms (25% faster).
- Shipped a CSV export feature covering 6 report types, enabling offline analysis without third-party tools.

URL Shortener API

Node.js, Express.js, MongoDB, REST API

GitHub

- Launched a shortening service handling 1,000+ URLs with collision-resistant Base62 hashing and TTL-based expiry.
- Added a B-tree index on the slug field, dropping redirect lookup from ~50ms to ~35ms — a 30% gain under load.
- Covered 8 edge-case scenarios (expired, malformed, missing slugs) with structured error codes and descriptive HTTP responses.

Blockchain Voting System

React.js, Express.js, MongoDB, Solidity, Ethereum, MetaMask, Web3.js

GitHub

- Wrote 3 Solidity smart contracts governing candidate registration, vote submission, and tamper-proof on-chain tallying.
- Enforced a single-vote constraint at the contract level via MetaMask wallet identity, making duplicate voting cryptographically impossible.
- Wired 2 event listeners to the chain to push live vote counts to the UI with zero polling overhead.

EXPERIENCE

Independent Full-Stack Developer

2024 – Present

- Shipped 6+ standalone modules spanning authentication, data pipelines, and client-facing dashboards using Node.js, Express, and MongoDB.
- Profiled and fixed 15+ API bottlenecks through query analysis and caching, achieving sub-100ms response times on critical routes.

ACHIEVEMENTS & ACTIVITIES

- Cleared 300+ DSA challenges on LeetCode and CodeChef, specializing in dynamic programming, graphs, and sliding-window techniques.

RELEVANT COURSEWORK

Data Structures & Algorithms · Object-Oriented Programming · Operating Systems · Database Management Systems · Computer Networks